

RISK REGISTER

ID	Portfolio / Application / Service	Threat Description	Risk Description	Owner	Availability Impact	Confidentiality Impact	Integrity Impact	Probability	Risk Score	Mitigation Details	Residual Risk	Risk Treatment / Notes	
R1	Fraud Detection System	Dataset too large for efficient processing	Slow data processing and model training	Anderson (Data Engineer)	Medium	Low	Medium	Medium	High	Use sampling; optimise transformations; run heavy tasks on Colab/labs	Medium	Monitor performance throughout project	
R2	CV Module	Limited forged-document examples	CV model underperforms or misclassifies	Jibola (CV/NLP Lead)	Low	Medium	Medium	Medium	Medium	Use lightweight CV model; generate synthetic forgeries	Low	Keep CV scope light to match project timeline	
R3	ML Models	Over-sensitivity to rare patterns	High false positives → unnecessary alerts	Gaurav (ML Engineer)	Medium	Low	Medium	Medium	High	Threshold tuning; SHAP analysis to refine model behaviour	Medium	Weekly monitoring of FP rate	
R4	Backend (FastAPI)	Low performance API routes	Real-time latency becomes too high	Aniruddha (Backend Engineer)	High	Low	Medium	Medium	High	Use async code, caching, and lighter models	Medium	Conduct load testing regularly	
R5	Dashboard (Frontend)	Poor rendering optimisation	Dashboard becomes slow	Onyinye (Frontend Engineer)	Medium	Low	Low	Low	Medium	Limit rerenders; optimise React states; test often	Low	Track FPS and responsiveness	
R6	Integration	Backend/frontend/model mismatch	System fails during integration	Entire Team	High	Medium	High	Medium	High	Early integration tests; API documentation	Medium	Dedicated integration day	
R7	Project Team	Uneven distribution of tasks	Workload imbalance affects output	Jibola (Product Manager)	Low	Low	Low	Medium	Medium	Weekly review; follow RACI closely	Low	Redistribute tasks if required	
R8	Project Timeline	Delays in deliverables	Deadlines risk being missed	Jibola	High	Low	Medium	Low	High	Use GitHub board; track progress weekly	Medium	Break work into smaller tasks	
R9	ML Fairness	Behaviour-based bias	Ethical blind spots affecting decisions	Gaurav + Jibola	Low	Medium	Medium	Medium	Medium	Fairness checks; subgroup evaluation	Low	Include in Ethics section	
R10	CV + NLP Models	Too much complexity added	System becomes harder to integrate & test	Aniruddha + Jibola	Medium	Medium	Medium	Low	Medium	Use smaller models; focus on core workflow	Low	Limit multimodal scope to essentials	